AMOD−N passive RF and analog blocks

Jason Gallicchio and Claude, rev A, 2026−10−25 github.com/gallicchio/AMOD−and−RFMOD

FOOT

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n9 Oscillator
Pmod 3.3 V
slide up: on
TUNE +−2 V
OUT −22 dBm

12 to 9 MHz

−2V +2V

n1 Calibration
SHORT, OPEN,
LOAD at the pin
25 OHM: VSWR 2
THRU 115 ps
tiles measure
from their pads

github.com/gallicchio/
AMOD−and−RFMOD
handbook and generator

n3 Transformer splitter
0.5−400 MHz
−3.2 dB each way
isolation 25 dB
a combiner too

isolation 0 to −40 dB

500k 400M

n2 Resistive splitter
DC−6 GHz
−6 dB each way
no isolation
IN max +20 dBm

n7 Bias−T
DC 16 V max
100 mA max
RF+DC has DC:
mind its load

n6 Detector
IN < +10 dBm
OUT to a
HIGH−Z ADC
square law
below −20 dBm

n5 Low−pass
DC−1 GHz
−3 dB 1.3 GHz
−30 dB 1.9−5 G

0 to −60 dB

10M 5.5G

0 to −40 dB

0.1G 6G

n4 Stub
notch 2.45 GHz
and 7.35 GHz
50 ohm

n11 Transformer
3−800 MHz
OUT: 2x V,
4x the Z
1: 200, 2: 49.9

n10 Balun
OUT floats
0.4−500 MHz
0.25 W, 30 mA

n8 Attenuator
NEEDS THE PMOD'S 3.3 V
by hand: SPI off, bits on
FPGA: bits off, SPI on
IN max +23 dBm, +9.5 dBm
below 0.5 MHz

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AMOD−N: passive RF and analog blocks. IN is each tile's top−left jack, OUT its bottom−right. No USB: n8 and n9 run on the Pmod's 3.3 V when it is plugged in; everything else needs nothing. See the handbook.

AMOD-N rev A, bottom side seen from below (mirrored), scale 1.90:1